

THE HARMONIC QUADRUPOLE OSCILLATOR AND ITS ANHARMONIC MULTIPOLE DESCENDANTS

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The Schrödinger equation with the noncentral local confining potentials $\sum_{m=1}^M \sum_{n=0}^m g_{mn} r^{2m} \cos^{2n} \vartheta$ is solvable in terms of matrix continued fractions. This is illustrated on the simplest example for $M = 1$.

A peculiar property of the anisotropic harmonic oscillator

$$-(\hbar^2/2\mu)\Delta\psi(r) + V(r)\psi(r) = E\psi(r), \quad (1)$$

e.g., with the quadrupole potential

$$\begin{aligned} V_Q(r) &= r^2(g_0 + g_1 \cos^2 \vartheta) \\ &= g_0 x_1^2 + g_0 x_2^2 + (g_0 + g_1)x_3^2, \end{aligned} \quad (2)$$

is its exact solvability in cartesian coordinates by the separable ansatz

$$\psi(r) = \psi_m(x_1)\psi_n(x_2)\psi_k(x_3)$$

(cf. any textbook). This property is completely lost when anharmonic corrections are added, e.g.,

$$V_{AQ}(r) = (r^2 + \gamma r^4)(g_0 + g_1 \cos^2 \vartheta). \quad (3)$$

As far as we know, only the special (isotropic) case $g_1 = 0$ of $V_{AQ}(r)$ again permits the separation of variables in eq. (1).

Considering some particle enclosed within an anisotropic (crystalline, etc.) confining medium, the rotational asymmetry of $V_{AQ}(r)$ may prove to be of the same importance as the vibrational anharmonicities. Hence, we shall describe here a method adequate for the treatment of the class of nonseparable problems like (1) + (3).

The main idea of the method [1] is to transform the two-dimensional partial differential equation (1) into the matrix form

$$(Q - E)D = 0, \quad (4)$$

in some adequate basis, and to employ fully the sparse structure of the infinite matrix Q . In fact, this is just a slight modification of the matrix continued fraction (MCF) method applied to the anharmonic oscillator [$V_{AQ}(r)$ with $g_1 = 0$] by Graffi and Grecchi [2].

The present method will presumably be applicable to all the potentials of the form

$$V_M(r) = \sum_{m=1}^M r^{2m} \sum_{n=0}^m g_{mn} \cos^{2n} \vartheta. \quad (5)$$

To simplify the discussion as much as possible, we shall describe here in detail just the first nontrivial case $M = 1$. It will have two advantages:

(1) The generalization to any $M > 1$ is straightforward.

(2) Any predictions may be tested because $V_1(r)$ coincides with the exactly solvable potential $V_Q(r)$ (2). The scaling of the radial variable $r \rightarrow \rho r$ enables us to require also $g_{10} + g_{11} = 1$, i.e., $g_{10} = 1 - g$, $g_{11} = g$.

Because of the broken rotational symmetry of eq. (2), it is sufficient to search for solutions of eqs. (1) + (2) which are axially symmetric and possess definite parity, in the form of a partial wave expansion

$$\psi_{\pm}(r) = \sum_{l=0}^{\infty} \frac{1}{2} [1 \pm (-1)^l] \psi_l(r) Y_{l0}(\vartheta, \varphi), \quad (6)$$

in spherical harmonics [3]. In the letter, we shall show that a closed form of this solution may be given in terms of MCF [1,2].

For simplicity, we shall consider just the even-parity case [$\psi(r) = \psi_+(r)$, $l = \text{even}$]. Inserting eq. (6) into eq. (1), we get the infinite coupled set

$$\begin{pmatrix} H_0 + \alpha_0 r^2 - E, & \beta_2 r^2, & 0, & \dots \\ \beta_2 r^2, & H_2 + \alpha_2 r^2 - E, & \beta_4 r^2, & 0, \dots \\ 0, & \beta_4 r^2, & \dots & \\ \dots & & & \end{pmatrix} \begin{pmatrix} \psi_0(r) \\ \psi_2(r) \\ \psi_4(r) \\ \dots \end{pmatrix} = 0 \quad (7)$$

of ordinary differential equations. Here, $\hbar = 2\mu = 1$ and

$$H_l = -d^2/dr^2 - (2/r)d/dr + l(l+1)/r^2 + r^2$$

are the standard linear harmonic oscillator (LHO) partial-wave hamiltonians. The values of the constants α_l and β_l are

$$\alpha_l = -\frac{1}{2}g(l^2 + l - 1)/(l - 1/2)(l + 3/2),$$

$$\beta_l = \frac{1}{4}g l(l - 1)/(l - 1/2)[(l - 3/2)(l + 1/2)]^{1/2},$$

and follow from the straightforward evaluation of the matrix elements $\langle Y_{l0} | \cos^2 \vartheta | Y_{l'0} \rangle$.

In the origin ($r \rightarrow 0$), the regular behaviour of the wave function components

$$\psi_l(r) = \text{const. } r^l [1 + O(r^2)],$$

is consistent with the system (7) since the coupling [off-diagonal components of the left-hand-side operator matrix $\mathcal{H}$ in eq. (7)] may be neglected. In the asymptotic region ($r \rightarrow \infty$), the different coupled components $\psi_l(r)$ are of the same order of magnitude. Fortunately, outside a finite subspace with $l \leq l_0$, the dominant parts of $\psi_l(r) \sim \exp(-\lambda_l r^2/2)$ become roughly equivalent,

$$\lambda_l = (1 + \alpha_l + \beta_l + \beta_{l+2})^{1/2} = 1 + O(1/l).$$

Hence, we are permitted to use the ansatz

$$\psi_l(r) = \sum_{n=0}^{\infty} R_{nl}(r) d_{nl}, \quad (8)$$

where $R_{nl}(r)$ denotes the orthonormalized LHO wave functions [3]

$$R_{nl}(r) = (-1)^n \left[\frac{2n!}{\Gamma(n + l + 3/2)} \right]^{1/2} \times r^l e^{-r^2/2} L_n^{l+1/2}(r^2),$$

with the oscillator parameter $\lambda = 1$.

It is a matter of simple algebra to show that the insertion of eq. (8) into eq. (7) leads to the infinite set of the linear algebraic equations for the unknown expansion coefficients d_{nl} ,

$$\begin{pmatrix} A_1 - E & B_1 & 0 & 0 & \dots \\ B_1^T & A_2 - E & B_2 & 0 & \dots \\ 0 & B_2^T & A_3 - E & B_3 & \dots \\ \dots & & & & \end{pmatrix} \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ \dots \end{pmatrix} = 0, \quad (9)$$

where D_k , A_k and B_k are $k \times 1$, $k \times k$ and $k \times (k + 1)$ matrices, respectively, with the nonzero elements

$$D_k^i = d_{k-i, 2i-2},$$

$$A_k^{ii} = (2k - 1/2)(2 + \alpha_{2i-2}),$$

$$A_k^{i+1i} = A_k^{ii+1} = 2\beta_{2i}[(k-i)(k+i-1/2)]^{1/2},$$

$$i \neq k,$$

$$B_k^{ii-1} = \beta_{2i-2}[(k-i+1)(k-i+2)]^{1/2},$$

$$i \neq 1,$$

$$B_k^{ii} = \alpha_{2i-2}[(k-i+1)(k+i-1/2)]^{1/2},$$

$$B_k^{i+1i} = \beta_{2i}[(k+i+1/2)(k+i-1/2)]^{1/2}$$

$$i \neq k,$$

$$i = 1, 2, \dots, k, \quad k = 1, 2, \dots$$

The infinite-dimensional matrix Q in eq. (9) has a block-three-diagonal structure. It was shown in ref. [1] that the complete solution of such a system may, under some general assumptions, be obtained in compact form when the concept of the MCF is introduced. In the full analogy with classical continued fractions [4] e.g.,

$$f_k = -\frac{1}{a_k - b_k^2 \frac{1}{a_{k+1} - \dots}},$$

where $a_k, b_k, \dots$ are some numbers (1×1 matrices), the MCF's $F_k (= k \times k$ matrices), $k = 1, 2, \dots$ will also be generated by their finite approximants ($F_{k,N}, k = 1, 2, \dots, N, F_{N+1,N} = \dots = 0$) in the limit $N \rightarrow \infty$. In the present context, their definition is given in the form of the matrix recurrences

$$F_k = \frac{1}{E - A_k - B_k F_{k+1} B_k^T} \quad (11)$$

In accordance with ref. [1] the Green's function of our problem coincides with the scalar quantity $F_1 = F_1(E)$. Its poles determine the eigenvalues so that the secular equation reads $1/F_1(E) = 0$, i.e.,

$$E = A_1^{11} + \sum_{m,n=1}^2 B_1^{1m} F_2^{mn}(E) B_1^{1n} \quad (12)$$

When recalling some recent applications [5] of classical continued fractions, our MCF precisely preserve the analogy. Alternatively, condition (12) may be interpreted [1] as the zero-determinant requirement in the homogeneous matrix equation (9). Finally, the formal representation of the eigenstate components $\psi_l(r)$ may be written now with the coefficients

$$d_{nl} = D_{n+(l+2)/2}^{(l+2)/2}.$$

These quantities are defined [1] step by step,

$$D_1^1 = 1, \quad D_k^m = \sum_{i=1}^{k-1} \sum_{j=1}^k D_{k-1}^{ij} B_{k-1}^{ij} F_k^{im}(E), \quad (13)$$

$$m = 1, 2, \dots, k, \quad k = 2, 3, \dots,$$

in terms of the MCF's $F_k(E)$ with E equal to the root of eq. (12).

In ref. [5], the classical continued fraction formulas throw new light on the convergence problems in perturbation theory. Due to the smallness of the off-diagonal matrix elements Q on the left-hand-side of eq. (9) for the small quadrupole component in $V_Q(r)$, the perturbative character of the present MCF recurrences (11) in the limit $g \rightarrow 0$ is also obvious, at least for the first (ground) state with $E = E_1 = 3 - g + O(g^2)$. The rapid increase of the dominant diagonal elements of A_k , $k \rightarrow \infty$, guarantees also the convergence of MCF's (11) for small g . In a similar context, namely in the MCF representation of the Green's function for anharmonic oscillators [2] or for the generalized Eckart potential [6], the numerical use of

eq. (12) proved to be a surprisingly efficient practical algorithm, even for large "perturbation" [$g = O(1)$] or when compared with other specialized methods. Concerning typical results of the numerical tests and some further technical details, we may also quote here a somewhat related paper ref. [7].

When we consider excited states with $E = E_p, p > 1$, we expect that

$$E_p(1 + O(g)) = E_p^{\text{LHO}} = 2(2p - 1/2). \quad (14)$$

At the same time, rewriting eq. (12) in more detail,

$$E = 3 - g + g^2 \frac{E - 7 + g + O(g^2 F_3(E))}{(E - 7)^2 + 4g(E - 7) + 3g^2 + O(g^2 F_3(E))}, \quad (15)$$

we get $g^2 F_2(E) = O(1)$ for $p > 1$. Resolution of this apparent paradox is easy. When we imagine that $E - A_p^{ii}$ changes sign and is of order $O(g)$ for $E = E_p^{\text{LHO}} + O(g)$, we obtain an estimate $F_p(E) = O(g^{-p})$. Hence, the modification

$$E = 3 - g + W_1^{11}(E),$$

$$W_r(E) = B_r \frac{1}{E - A_{r+1} - W_{r+1}(E)} B_r^T, \quad (16)$$

$$r = 1, 2, \dots, p - 1,$$

$$W_p(E) = B_p F_{p+1}(E) B_p^T,$$

should replace eq. (12) when we intend to interpret it as a Brillouin-Wigner-like perturbation formula. It is worth mentioning that the effective interaction $W_1(E)$ in eq. (16) is a rational function of the matrix elements of $F_{p+1}(t)$ or $F_{p+2}(E)$ etc. Hence, explicit perturbation expansions may very efficiently be generated from eq. (16) when using a symbolic manipulation language (e.g., REDUCE) on the computer. Their rational form (location of poles) provides valuable information about the convergence of the standard perturbation theory.

In conclusion, we emphasize once more that the same MCF method applied to any potential $V_M(r)$ leads to analogous results. The only difference will appear in the elements and dimensions of the partitions in the block matrix Q in eq. (9). For example, we get $\dim A_k = \dim F_k(E) = 3k - 2$ for the anharmonic quadrupole potential $V_{AQ}(r)$ (3).

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